

Deterministic Optimization

Simplex Method in a Nutshell

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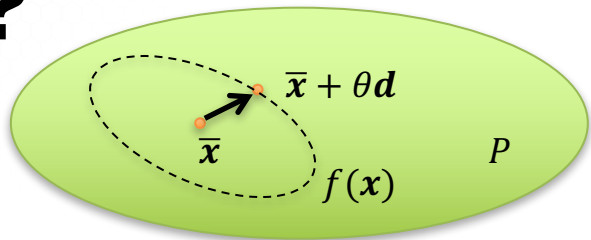
Local Search for LP

Simplex Method in a Nutshell

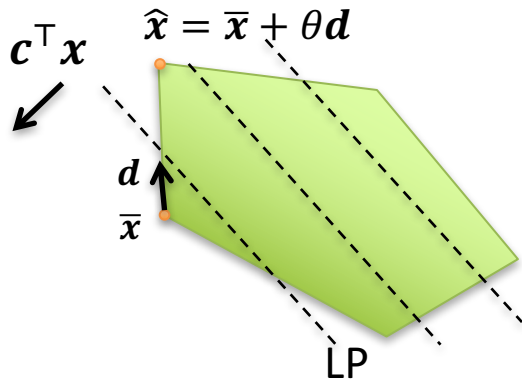
Learning Objectives for this module

- Discover how to apply local search to linear program

How to Apply Local Search to Solve LP?



General case



- For LP, objective function is linear, therefore, linear level sets
- an optimal solution, if exists, is an extreme point (BFS)
- We start the local search from a BFS $\bar{x}$, potentially not optimal
- Find a direction d to move to a point with lower objective value
- Since we only care about BFS, d should point to another BFS
- Find a proper step size θ so that $\bar{x} + \theta d$ is a new BFS

Key Steps to Implement Simplex Method

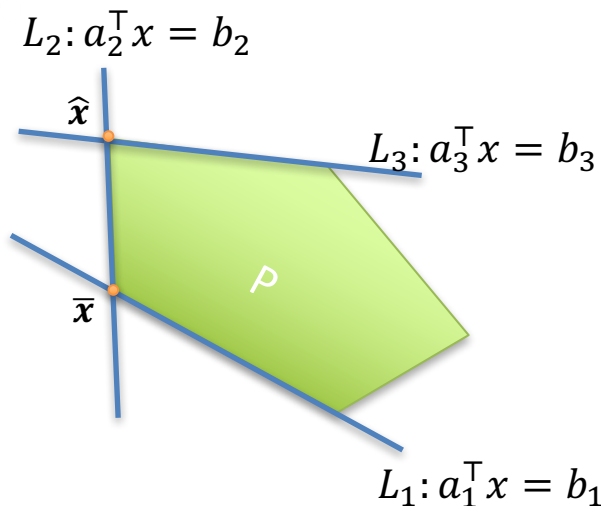
- The key step is to find a direction d and stepsize θ so that:
 1. d Points from a BFS to one of its adjacent BFS's
 2. That adjacent BFS should reduce objective value
 3. Move along the favorable direction as much as possible to maintain feasibility and to reduce objective
 4. Stop when optimal solution is found (or cannot be found)
- Let's focus on number 1 first:
 - Need to keep the next iterate feasible
 - Need to make the next iterate an adjacent BFS

Partition Vectors & Keep Feasibility

- A BFS x is partitioned into a sub-vector of basic variables x_B and sub-vector of nonbasic variables x_N :
 - $x = \begin{bmatrix} x_B \\ x_N \end{bmatrix}$
- We can also partition the direction vector d accordingly:
 - $d = \begin{bmatrix} d_B \\ d_N \end{bmatrix}$
- $x + \theta d$ should be **kept feasible**: $A(x + \theta d) = b$
 - Since $Ax = b$, and step size $\theta > 0$
 - We have an **important property of d** : $Ad = 0$
 - $[B, N] \begin{bmatrix} d_B \\ d_N \end{bmatrix} = 0 \Rightarrow Bd_B + Nd_N = 0$

An Adjacent BFS

- What does it mean for two BFS's to be **adjacent**?
 - Two BFS are adjacent if they share the same $n - 1$ linearly independent active constraints



Feasible region P :

$$a_1^T x \leq b_1$$

$$a_2^T x \leq b_2$$

$$a_3^T x \leq b_3$$

$$a_4^T x \leq b_4$$

$$a_5^T x \leq b_5$$

At $\bar{x}$, L_1 and L_2 are L.I.A.C.
At $\hat{x}$, L_2 and L_3 are L.I.A.C.
 $\bar{x}$ and $\hat{x}$ are adjacent BFS's
and they share 1 L.I.A.C. (L_2)

Adjacent BFS's in Standard Form LP

- In standard form LP with A is $m \times n$ matrix, x is $n \times 1$ vector

$$\min c^T x$$

$$\text{s.t. } Ax = b$$

$$x \geq 0$$

- Every BFS has to satisfy all m equality constraints $Ax = b$
- So two adjacent BFS's must share the same set of $n - m - 1$ nonbasic variables as $n - m - 1$ active constraints, and differ in one nonbasic variable
- Example, if $x = (x_1, \dots, x_5)$ has nonbasic variables $x_3 = x_4 = x_5 = 0$, then its adjacent BFS must share two of these three nonbasic variables.

Summary

- We learned:
 - The geometric picture of applying local search to solving LP
 - Key steps in implementing the geometric picture by an algorithm (simplex method)